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Use of different plant tubers for therapeutic purposes by the ethnic tribes of Khunti district Jharkhand, India

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Abstract : The present paper enumerates different tubers that are widely used as medicine by the ethnic tribes and other indigenous non-tribe communities of Khunti district of Jharkhand. During the ethno-medicinal field survey of Khunti district, altogether 26 species of plant tubers belonging to 22 genera and 11 families, growing in remote areas, which are used for curing various diseases like cough, asthma, diorrhoea, skin diseases piles and other common diseases have been identified. The present article report such tubers, their local names, botanical name, family, availability, and method of use for curing diseases. Some of the important such tubers are *Acorus calamus* (Ghora buch), *Amorphophallus paeoniifolius var. campanulatus* (Ol),*Chorophyton tuberosum* (Safed musli), *Coccinia grandis* (Van Kundru*), Curculigo orchioides* (Kali musli),*Curcuma decipiens*(Jangli haldi, Van Haldi), *Cyperus rotundus* (Nagarmotha), *Gloriosa superba* (Kalihari), *Pueraria tuberosa* (Patal Kohra), *Urginia indica* (Jangli piyaz).

Keywords: Ethno-medicinal,Horopaths,Pahans, Indigenous, Ethnic Tribes,Therapeutical.

INTRODUCTION

Khunti district of Jharkhand, primarily dominate with different tribal communities, is very rich in its phyto-diversity where enormous number of ethno-medicinally important plants grow either in remote forests or in waste lands and even on road sides. The district has more than 20 different ethnic tribe communities that constitute 72-73% of the total population of the district. These ethnic tribes are primarily nature loving and lives either in remote forests or in adjoining areas since time immemorial, maintaining their socio-cultural, socio-economic and socio-religious institutions intact. The most striking point is that this mass is still dependent on various forest produces, both major and minor, for their livelihood and subsistence, and hence is also called as Forest Dependent Population. The ethno-medicinal field survey of Khunti district was conducted during the years 2010-2012, and it was revealed that the ethnic tribes of this area are very expertise in utilizing different plant parts as medicine for the treatment of almost all sorts of diseases. They utilize different tubers as food and medicine for the treatment of different diseases. They know the nutritional and medicinal value of the tubers, which are not in use by the urban populace. This type of documentation of the medicinal properties of tubers has not yet been done in a systematic manner either by the Botanists nor the Anthropologists.

These ethnic tribes of the area go to the local horopaths, pahans, vaids, kabirajs and other local knowledgeable persons for their all sorts of pharmaceutical needs. The indigenous herbal medicine practitioners (Horopaths) treat all sorts of ailments with the indigenous plants. They use many tubers also for the treatment of diseases. In the present article documentation and enlistment of different tubers which are used as medicine by the ethnic tribes for the treatment of various ailments have been done.

MATERIALS AND METHODS—

The tubers used by different ethnic tribes of Khunti

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District to cure various ailments /diseases were identified during the ethno-medicinal field survey conducted during the years 2010-2012, for the fulfillment of the U.G.C. New-Delhi, Sponsored Minor Research Project, entitled "Survey and Inventorization of Medicinal Plants of Khunti District of Jharkhand, India". Collection of live forms of important ethno-medicinal plants and documentation of ethno-medicinal properties was done by conducting direct interview with the horopaths (indigenous ethnic herbal medicine practitioners), pahans, vaids, herbal medicinal plant collectors and other local knowledgeable persons of the study area, following the standard methods (Jain, S.K. 1963[1]; Jain, S.K.1964[2]; Jain, S.K. &D.C.Pal, 1976[3]; Jain, S.K. 1995t). Important plants tubers were collected and herbaria have been prepared & preserved in the Ethno-botanical Laboratory, University Department of Botany, Ranchi University, Ranchi. Documentation of all relevant information regarding the medicinal properties of the plants/ plant parts was done through direct interview with the local Horopaths, pahans, Kabirajs, vaids, and other knowledgeable persons of the area. Collection and documentation of data was done with the help of a questionnaire containing questions on different ethno-botanical and medicinal aspects. Help from local plant collectors and other indigenous knowledgeable persons were sought in selection of survey site and collection of plants.

RESULTS AND DISCUSSION –

Enlistment and inventorization of medicinal plants and the documentation of ethno-medicinal properties of such plants was done during the ethno-medicinal field survey of Khunti district in Jharkhand state in the years 2010-2012. It was revealed that the ethnic tribes of Khunti district are very expertise in utilizing various wild as well as cultivated plants as medicine for the treatment of almost all diseases. In course of the survey it was noticed that the ethnic tribes utilize different tubers wild as well as cultivated as medicine for the cure of various ailments/ diseases/disorders ranging from simple cold-cough, to fever, skin diseases, piles to wounds, to diorrhoea to asthma to different common diseases. Altogether 26 tuber plants have been enlisted belonging to 22 genera and 11 families that are in use as medicine**.** Such tuberous plants have been arranged alphabetically giving their botanical name, followed by family name in parenthesis, followed with local names/common names in capital letters and finally the ethno-medicinal uses of the tubers.

1**.** ***Acorus calamus*** L. " Araceae", GHORA BUTCH, BACH**; Ethnomedicinal uses** –**In sore throat** —Small piece of rhizomatous root is chewed & inhaled once or twice for 5-7 days. It gives relief in sore throat. **In cough and bronchitis** — small piece about 3-5 gms of rhizome is boiled in about 150 ml water for 10-12 minutes. A little salt is added. The decoction is inhaled twice daily for a week .It gives complete relief in cough and bronchitis. **As anthelmintic** — 3-5 gms of dry powder of rhizome is inhaled with warm water for 5-7 days in night before going to sleep. It kills the intestinal worms. The rhizome is used for the treatment of depression and other mental disorders. Rhizome is also given for diorrhoea & dysentery.It is useful in intermittent fever, skin diseases and general debility. The rhizome is also used in the treatment of cancer. **In Chronic diorrhoea**- The patient if given the paste of the tuber. Dose- One/two teaspoon paste of the tuber is inhaled orally with a glass of water twice a day for three-four days. It cures the diorrhoea.

2. Allium sativum Linn. " Liliaceae";LAHSUN;GARLIC. **Ethno-medicinal uses- In gastro-intestinal problems—** The bulb of *Allium sativum* is used as medicine for the treatment of various gastro-intestinal problems. **In hypertension**— The bulb is very effective in case of hypertension. It is very good for preventing hypertension. **In arteriosclerosis**— The bulb is also used in case of arterial hardening (arteriosclerosis). It is recommended to eat at least one bulb of garlic every day to stay in good health. **In heart & other ailments**— The bulb is very effective in controlling blood pressure and in different types of heart problems. The bulb is used as medicine for cough, whooping cough, bronchitis, asthma, facial paralysis, duodenal ulcers, hysteria, sciatica, swellings and in dental problems.

3. Allium cepa Linn. "Liliaceae";PIYAZ;ONION. Ethnomedicinal uses —The bulbs are used as a medicine in diarrhoea, dysentery, bronchitis, asthma. Bulbs are also used in leucoderma & skin diseases. Bulbs are useful in malarial fever, jaundice, tumors, wounds, paralysis and in controlling vomiting tendency. **In blood dysentery** – Bulbs of onion cut vertically, a little lime is added through the

slit and kept as it is for about 30 minutes. Then it is chewed directly for blood dysentery.

4. ***Alpinia galanga*** (Linn.)Willd. "Zingiberaceae"; KULANGAN; SUGANDMULA. **Ethnomedicinal uses – In Arthritis** – Rhizome is used for the remedy of arthritis. The rhizome is also used as a medicine for the treatment of asthma, bronchitis & cough. The rhizome is also given as a drug against intermittent fevers and stomach pain. The rhizome is also useful in diabetes, cough & tubercular glands.

5.Amorphophallus paeoniifolius var. ***campanulatus*** (Decne)Sivad;***A. bulbiferi L***. ."Araceae",Ol, Suran, Jamikand. Eng. – Elephant foot yam. **Ethno-medicinal uses- In piles**- Dry powder of the tuber is made. One teaspoon of the dry powder is inhaled orally with about half glass of warm water twice daily (morning and evening) continuously for 15-20 days. It is to be repeated after 15 days if the problem exists. **In wounds and burns**- Dry powder of the tuber is made. Dry powder of the tuber is mixed with coconut oil in 2:1 ratio and mixed nicely. The paste so prepared is applied on the wounds and burns twice or thrice daily for 3-5 days in wounds and for 15-20 days in case of burns. The corm is used for cough, bronchitis and asthma. The corm is also used as medicine in anaemia, amenorrhoea, helminthiasis, constipation, tumors, seminal weakness, vomiting, anorexia and general debility. The roots are used in opthalmia, amenorrhoea & boils.

6. Asparagus racemosus Willd. Fam. "Liliaceae". Satavar, Satmuli. Eng. Wild asparagus. **Ethno-medicinal uses** – The whole plant is good tonic considered most suitable diet for the recovering patients and post- natal women. **For fever**- Roots are used as medicine for fever. Dose— Root paste is made. Two tea-spoons twice daily (morning & evening) for 3-5 days are given to cure fever. For children between 5- 10 years 1-Tea-spoon is advised.25gms of tuberous root with misri twice daily for 8-10 days taken. It cures fever. As Medicine for various gynecological problems – a)During delivery- About 2-teaspoons of the root paste is orally administered during labor- pain. The dose is repeated every 2-hours. It facilitates delivery. b)Sometimes, root paste is applied on the Vagina during labor pain. It helps in smooth delivery. c) Tuberous root of *Asparagus racemosus* mixed with equal amount of root of *Hibiscus rosa-sinensis* & rhizome of *Bombax ceiba* (Semal) are ground to powder. One teaspoon of the powder in one glass of milk is taken daily in the morning in empty stomach continuously for 8-10 days. It cures from white or yellowish discharge from the vagina (Leucorrhoea). It cures backache & weakness in the body. d) **For menstruation problems** — Asparagus root mixed with equal amount of rhizome of *Musa paradisica* (kela) & flowers of *Curcuma longa* (haldi) are made to paste. One spoon full of the paste mixed with one glass of fresh unboiled cow milk taken in empty stomach early morning continuously for 7-9 days. It is used for the treatment of various menstruation problems. e) Many women suffer from weakness, giddiness, loss of appetite and sometimes loose motion during different stages of pregnancy or after delivery. It happens due to haemoglobin deficiency. The body looks yellowish. The disease is called as Anaemia. The local herbal medicine practitioners give root powder of *Asparagus racemosus* as a tonic cum medicine for such women. Dose—One tea spoon full of A. racemosus with one glass milk to be taken continuously during pregnancy. The dose if taken after pregnancy for at least 20-25 days. f) Root powder of *A. racemosus* is also used by women after delivery. Root powder cooked with milk & given to such women for 25-30 days after delivery. It increases the milk secretion to a considerable extent. In chest pain —Rhizome of *A. racemosus* grinded with *Curcuma longa* (Turmeric) & filtered. The filtrate is administered for chest pain and stomach pain. Dose – two teaspoons full twice a day for 3-5 days. **In dehydration** — The rhizomatous tubers are chewed to check dehydration.

7. Ceropegia bulbosa Roxb. " Asclepiadaceae "; ***Cerogagia tuberosa*** Roxb. PATAL TUMBI;

KHAPPER KADDU; JANGLI SAKARKAND, GAKERKUND; KEWDA KI NAL; The tuberous roots are used as medicine. The tubers are bitter in taste. The bitterness is removed by boiling the tubers. The bitterness is due to an alkaloid Cerogagine.The tuber is rich in starch. It contains about 42-45% starch. **Ethno-medicinal uses— In gastro-intestinal problems**- The tuber is very effective in the treatment of various gastro-intestinal problems like indigestion, dysentery, diorrhoea, stomach pain etc. 10-15 gm of the tuberous root powder is inhaled orally with half glass of hot cow milk twice/thrice daily

for 2-3 days. **In diorrhoea and dysentery** – Fresh tubers are boiled. Two teaspoon of the boiled tuber paste is eaten twice daily for 3-5 days. It completely cures diorrhoea and dysentery. **In poisonous animal bite**- The powder of the tuberous root is applied externally on the affected area. Application is done twice thrice daily for 4-5 days. It checks the spread of the poisonous effect on other areas of the body and prevents the poisonous effect of the animal. **In urinary bladder stone**- Tuber decoction is administered orally for the treatment of urinary bladder stone. Two tea-spoon full tuber decoction is inhaled orally 2-3 times daily for 15-20 days. It gives rid of urinary bladder stone. **In deafness**- The seed paste is used. Seed paste is dropped in the ear in case of deafness. The paste if given once in the morning and once in the evening for a month.

8. Coccinia grandis L. " Cucurbitaceae"; VAN KUNDRU;RAKTAPHALA; ***Coccinia indica*** Wight and Arn ; ***Coccinia cordifolia*** Cogn.***Cephalandra indica*** Naud. **Ethno-medicinal uses- In fever, cold and cough** - One teaspoon powder are inhaled orally with half glass hot water twice /thrice daily for 3-5 days. It gives complete relief. **In wounds, ulcers, stomatitis**- The paste of the root is applied on the wounds and ulcers. 2-3 times daily continuously for 7-9 days gives complete relief from wound and ulcers. **In skin diseases**- Root paste is applied externally on the affected skin regularly for 10-15 days. Application is done 2-3 times daily. **In different stomach problems**—One teaspoon root powder is inhaled orally twice or thrice daily for 5-7 days. It gives relief from different stomachial problems.

9. Chlorophyton tuberosum (Roxb) ; " Liliaceae"; SAFED MUSLI; **Ethno-medicinal uses- In seminal debility**- It is used for the treatment of problem related to seminal debility. One and half teaspoon of the dry powder of cladodes is taken orally with one glass milk once daily for 15-20 days. The dose is to be repeated after one month, but with one teaspoon of cladode this time. **As a tonic** – Half teaspoon of dry tuber root powder is mixed with half glass of water and inhaled orally twice daily. It gives strength vigor and is a very effective tonic for male. **In leucorrhoea**- 2-3gm of tuber powder is administered orally twice or thrice daily for 5-7 days during the period. It gives relief to a greater extent from leucorrhoea.**In fracture of animal bone**—The root powder as paste is applied in the fractured bone in animals. The root powder is also given orally to the animal in case of bone fracture. The medication is continuously done for 15-20days.

10. Costus speciosus Koen.ex. Retz. Smith. "Zingiberaceae".KITAMULI; Keokand; CHANDA; MAHALAKARI. ***Ethno-medicinal uses*** – Rhizome powder boiled with milk and sugar is taken at night for 15 days as a tonic. Rhizome is made into paste and applied on joints in rheumatism. Two teaspoon dry rhizome powder is inhaled orally twice daily for 7-9 days to cure rheumatism. Two tea spoons of extract of rhizome are given once daily for seven days to relieve pain in bone. Two tea spoonful powder of rhizome is taken orally against snake bite. The rhizome is edible after cooking. It is rich in starch. For increasing milk secretion – The rural aboriginal people use to give warm extract of the rhizome to the new mothers after delivery. It increases milk secretion. Dose – 2-3 tea spoons full of warm rhizome extract two three times a day for 5-8 days after delivery. Rhizome paste is applied on head and also administered orally for cooling effect.Dose—2 tea spoons once a day for 3-5 days. In gastric troubles—Rhizome extract is also used for gastric troubles. 2-3 teaspoons once a day for 4-7 days. In giddiness – Rhizome of *C. speciosus* +Termeric *(Curcuma longa*) in equal amounts mixed and grounded and the paste made into pills. The pills are used for giddiness. The rhizomes are also used in skin diseases. In Asthma- Rhizome infusion is given for asthma. In ear ache – 3-4 drops of rhizome decoction is poured in the ear twice daily for5-7 days for curing ear ache. In skin diseases- Rhizome paste is applied externally over the infected skin twice daily for 15-20 days .It cure skin diseases.

11. ***Colocasia esculenta***;(Linn.),Schott; ***Colocasia antiquorum*** Schott.Arvi; "Araceae";TARO; ARBI; KACHCHU. **Ethno-medicinal uses-** Tuber is used against superficial injuries. Tuber paste is applied as paint over the injured area for 3-5 days. In inflammation of joints and tears in the joints—The tubers are used as medicine for the treatment of arthritis and bruises. The method of treatment for external injuries, arthritis and sprains is as follow —The tuber is crushed and mixed with equal amount of wheat flour. To it is added a bit of grated ginger

and a little salt. The whole is mixed nicely. About 1cm thick patch of the mixed paste is made using thin cloth. It is then put on the affected area of the skin until it dries. The process is repeated for 5-7 days. In tonsillitis and nervous system- the tubers are also used against tonsillitis and as a nerve tonic. Kachchu is rich in fiber and it contains iron, calcium & potassium that help in treating tonsillitis and in nervous system.As hair tonic- The local horopaths advice to rub tuber juice of Colocasia on the scalp for good growth of hair. It serves as a hair tonic. It also checks the hair falling.

12. Curcuma decipiens Hook. "Zingiberaceae";***Curcuma amada*** Roxb.***Curcuma angustifolia*** Roxb. JANGLI HALDI;VAN HALDI;AMAHALDI;DARU HALDI;MANGA GINGER.**Ethno-medicinal uses—In skin diseases**- Curcuma is a very effective medicine for skin diseases. It is used as powder mixed with kala jeera and applied externally on the affected skin. The treatment is given twice daily for 8-10 days. If complete cure is not done then the application of drug is extended for further 5-7 days. **In jaundice-** The rhizome is very effective medicine for jaundice. The juice of the rhizome is inhaled orally with boiled water. About 2-3 teaspoon juice is taken with half glass boiled water twice daily for 8-10 days and then 1teaspoon once daily for further 7 days. It gives complete relief from jaundice. In dropsy-(In excessive accumulation of clear or watery fluid in any of the tissues or cavities of the body) – About 10 ml Rhizome decoction is inhaled orally 2-3 times daily for 10-15 days to cure dropsy. **To deactivate poisonous effect in snake or scorpion bite**- Rhizome powder is applied locally over the affected portion in case of snake or other poisonous insect bite. It prevents the poisonous effect.

13. Curcuma longa; "Zingiberaceae"; HALDI; TURMERIC; ***Ethno-medicinal uses***- **In diorrhoea** - Half glass hot water extract of the dried rhizome twice daily for 5-7 days is recommended for diorrhoea. **In diabetes**- One glass extract of rhizome is recommended to be taken orally for one month, once daily, in case of diabetes. The dose is reduced to half after a month. **In liver problems**- Half glass extract is inhaled orally continuously for 15-20 days in case of liver problems.**As a tonic**- The extract is also recommended as a tonic and to keep the stomach calm and cool.**In stomach disorders**- The horopaths recommend taking fresh juice of the rhizome regularly to keep the body away from different stomach disorders. Half to one glass fresh juice to be taken daily in the morning in empty stomach. **In promoting abortion** – Oral administration of the hot water extract of rhizome is recommended in promoting abortion. Pregnant women are not advised for oral administration of hot water extract of rhizome. Paste of the rhizome extract is used as a paint over the vagina for promoting abortion. **In food poisoning** — About 5 gm. of the rhizome-paste is eaten to treat food poisoning. It may or may not cause vomiting, but causes the restoration of the normal condition. Generally, 2-3 doses are enough to obtain the desired effect, each at an interval of 2 hours. About 5 gm. of the rhizome-paste is consumed with cow's milk, 2-3 times, each at an interval of 2-3 hours, to treat food-poisoning. **In body pain**- About 5-7 gm. of the rhizome-paste, taken early in the morning with honey or milk, is useful against body aches. The rhizome is an appetizer and also a very good restorative drug. It is to be continued for a week, at least or for as many days to bring out the desired effects. **In boils—** Rhizome-paste is applied over boil to dry and cure it. **In wounds** —Dry rhizome-powder is mixed with coconut oil to make a paste which is applied on wounds for the antiseptic and curative effects. More particularly, it is used on livestock. **As a milk purifier**- Dried rhizome is crushed to powder and halwa is prepared with ghee and gur or sugar. It is called as "SONTH" by local people. Two teaspoon of the sonth is given twice or thrice for 5-10 days to the mothers 2-3 days after delivery. It helps in milk purification and also increases moth's milk

14. Curculigo orchioides **Gaertn. "Amaryllidaceae"; KALIMUSLI, TALMULI (Kalimusli,Musli,)**

Ethno-medicinal Uses:-For asthma & jaundice – Extract from tuber is obtained. The extract is used for asthma and jaundice. Dose – 2-3 teaspoons two times daily till the patient is not cured. For cuts and wounds – Root paste is prepared by crushing the roots. The root paste is applied externally on cuts and wounds 2-3 times a day for about a week or till the recovery. Root stocks are sweet, cooling, bitter diuretic, carminative. In piles – Tuber extract is used for the remedy of piles. About 5ml of the root extract is given 2-3 times a day for about a week or even more till the recovery to the patients suffering from

piles. In scorpion sting — The roots are also used in scorpion sting. Root paste is applied to the affected portion and root extract is orally administered for at least 3-5 days, about 5ml of the extract is given in one dose twice daily. **In diorrhoea--** Rhizome is used as medicine for the treatment of diorrhoea. Dry rhizome powder is prepared .One teaspoon of the powder is inhaled orally twice daily for 3-5 days. It gives relief from diorrhoea.

***15. Cyperus rotundus* L.** Fam. Cyperaceae. Nagar motha, Motha. Eng. Nut grass.

Ethno—medicinal uses – In intestinal worms (Anthelmintic) — Rhizome of *Cyperus rotundus* and root of *Cynodon dactylon* are mixed in equal proportion, crushed and extracted the juice. Two -three teaspoon of the juice is taken 3-4 times with an interval of 3-4 hours to expel the intestinal worms. Rhizome paste is eaten for the anthelmintic action. About 20gm. of the paste is eaten in the first dose which is followed by that of 10gm. only. The course is 3 times a day for 2-3 days. However, the dose is reduced to half in case of children. Five-eight gm. of the rhizome is consumed in the morning for a week or more to increase the appetite. The rhizome is also useful in wounds, ulcers, fever, vomiting, malarial fever, diorrhoea, dysentery, skin diseases, scabies cough and general debility. In skin diseases and scabies- The tubers are used against skin diseases and scabies. One teaspoon paste is inhaled orally twice daily for a fortnight. It cures the skin diseases and gives relief from scabies also. In diorrhoea and dysentery – Tuber paste is given in diorrhoea and dysentery. 3-5 gm paste is inhaled orally with a glass of warm water 2-3 times a day for 3-5 days. It cures diorrhoea and gives relief from dysentery. In cough and bronchitis- Tuber of *Cyperus rotundus* is also given in cold, cough and bronchial problems. About 3-5 gm Tuber paste is taken 2-3 times a day for 5-7 days.

16. Dioscorea alata L."Dioscoriaceae";***D.globosa*** Roxb; ***D.rubella*** Roxb;CHUPRI ALU,AARU, YAM.

***Ethno-medicinal uses*—In fungal skin disease**— 4-5 gm paste of *D. alata* tuber is put on the infected part of the body and tied with thin cloth. It is kept for 6-8 hours and then removed. Again second time treatment is done after an interval of 5-6 hours. The process is repeated till complete recovery. **In piles**- One teaspoon tuber paste is inhaled orally twice-thrice daily for10-15 days. It gives relief in piles. **In gonorrhoea**—The tubers are used in gonorrhea also. One teaspoon dry tuber powder is administered orally 2-3 times a day for 20-25 days. It cures gonorrhea. **In leprosy**- Tuber paste is applied externally over the affected portion of the body as paint continuously for about a month or even more till cure. Dry tuber paste about 3-5 gm taken orally with one glass of water 2-3times daily during the same period.

17. Dioscorea esculenta Lour. ; "Dioscoriaceae";***D. Hirsuta*** Dennst.; ***D. hispida*** Dennst.;

D. pentaphylla L. PINDI, PINDALU; JATASHANKARI; BHUSA ; KAND;AMARKAND.**Ethno-medicinal uses**- The tuber of *D. esculenta* is used to increase weight in case of weight loss. One boiled tuber in the morning and one in the evening is administered orally continuously for 15-20 days. I t helps increase in weight. **In poisoning arrows**- Milky juice of tuber is used to poison arrows. **As a contraceptive**—The tuber is used as a contraceptive by the indigenous tribal and non-tribal ladies. 10-12 gm dry tuber powder is giver once in the morning continuously for five days after menses. **In abdominal pain**- One teaspoon dry tuber powder is given orally twice daily with water for cure of abdominal pain after delivery. **In asthma and cold/cough**—One teaspoon tuber decoction is administered orally 2-3 times a day for 5-7 days when the patient feels asthmatic problem or suffering from cold/ cough. **In snake bites**- The tuberous roots are used against snake bite. One glass root is crushed and mixed with four glass water. It is boiled till the water comes to one glass. It is cooled and inhaled orally – one glass at a time thrice daily for 3-4 days.

18. Gloriosa superba Linn. "Liliaceae"; KALIHARI; DAKSHNAG; MALABAR GLORY. **Ethno-medicinal uses—In skin diseases**- The tuber of kalihari is given for the treatment of skin diseases. Paste of tuber is applied externally as paint on the infected areas twice daily, in morning and in evening, for 8-10 days. It cures different skin infections. Rhizome in small quantity is used for vitality & strength. **In gonorrhea**— The rhizome is crushed & deep in water for 10-15 hours. Thick paste like substance (starch) accumulates at the base. This is used for the treatment of Gonorrhoea. This rhizome starch is applied internally. Dose – 1-2 ratti at a time, for 5-7 days and twice daily. It cures Gonorrhoea. **In snake/scorpion**

bite— The rhizome paste is applied externally in snake-bite & scorpion –bite. It gives relief from pain. The rhizome starch is used in very small quantity (1-2 ratti) in promoting labor pain and expulsion of the placenta. **In abortion** —The rhizome extract is administered orally for abortion. The rhizome paste with turmeric paste applied for 3-5 days in the abdomen for the treatment of Pirhi disease in children. The rhizome paste is applied on the body for chicken –pox. It was informed by the local herbal medicine practitioners that the rhizome paste should not be applied in big amounts, because it is highly poisonous and causes vomiting, purging and burning sensation

19 .***Ipomoea batatus (L.) Lamk***. "Convolvulaceae". MITHA ALU; SAKARKAND; SWEET POTATO. **Ethnomedicinal uses** –Tuberous roots are eaten. It is either boiled in water or directly roasted in fire. Some eat the boiled tubers after mixing with milk. It is a very common & cheap diet for the rural aboriginal people of Khunti district. Tuberous root is used as a medicine for the treatment of patients who suffers from frequent painful passage of urine (Strangury).The tuberous roots are also useful in constipation, burning sensation, diabetes and general weakness.

20.Ipomoea digitata (Linn.)Lam. ***I. paniculata*** R. Br. ***I. mauritiana*** Jacq. "Convolvulaceae". BHILAIKANDA;BHUMIKUMHARA;BHUIKONDAHA; GIANT POTATO. **Ethno-medicinal uses—As a tonic**— The tuberous root is used as a tonic. **As a aphrodisiac** – The tuberous root is used as a drug to stimulate sexul desire. **In emaciation of children**—One teaspoon powdered tuberous root is given with milk 2-3 times daily for longer period about a month or even more to weak children. It is very useful in emaciation of children. It helps in increasing strength and energy in children. **In indigestion** – The tuber root powder is also used as a medicine in case of poor digestion. **In sexually transmitted diseases**- the tuberous root is used in the treatment of syphilis and gonorrhea.

21. Momordica dioica Roxb.ex. Willd. "Cucurbitaceae";KHAKSA; BAN KARELA; KAKODA. ***Ethno-medicinal Uses*** – **For rib- muscular pains** – Tuberous root crushed with the juice of *Tamarindus indica* and the decoction is used as a medicine for the treatment of Rib-muscular pains. Dose – 2 tea spoons twice a day till recovery. **For dandruff** —-Tuberous root collected crushed &made to paste. It is applied on head. It removes dandruff from hair. **For Burns** — Leaf juice or leaf paste is very useful for Burns. It is prescribed by the local indigenous herbal medicine practitioners. **As insecticide & Anthelmintic** — The leaves have high insecticidal property. The acetone extract of the leaf has been proved to be toxic to the insect *Euproctis lunata.* The fruits are eaten as vegetable. The fruits are very good medicine for many diseases. **In weakness -** Paste of the dry powder is made mixing with water. Two teaspoon paste is inhaled twice daily, in morning and in the evening, continuously for a month. It is very beneficial in case of weakness.

22.Pueraria tuberosa (Willd) DC. "Papilionaceae"; "Fabaceae" PATAL KOHRA; VIDARI KAND. GHORA BEL. ***Ethno-medicinal uses–***In malarial fever- Root is ground with the water and taken orally in malarial fever,. One teaspoon 2-3 times a day for 5-7 days gives relief from malarial fever.In abortion- About 20ml root decoction is given 2-3 times for 3-5 days to pregnant women. It results in abortion. In boils and ulcers - The tuber is used on boils and ulcers.The tuber and other parts of the plant are used in body swellings. The tuber is pounded in water and then used in body swellings. As an anti nematode— About half spoon tuber paste is given to patient orally. It helps in easy expulsion of guinea worm. The tuberous root is useful in burning sensation, constipation, arthritis, cardiac debility, intermittent fever, leprosy, tuberculosis, cough, spermatorrhoea, dyspepsia and general debility. The tube is used as anti-inflammatory drug. The tuber is crushed to paste. The paste is applied externally on the swollen portion 2-3 times daily for 8-10 days.

23. Smilax zeylanica L. S indica N. Burm. Smilax ovalifolia. **Fam. Liliaceae. RAMDATWAN;JANGLI AUSHBAH;CHOBCNINI**J; INDIAN SMALAX. ***Ethno-medicinal uses –*** Bark decoction is used for gargling when the gums are spongy and bleeding. Bark is used in the treatment of diorrhoea, dysentery & liver complaints. The bark is also used to treat haemorrhagia, eye diseases, leucorrhoea, wounds, ulcers, tumors, leprosy, skin diseases, asthma, bronchitis, menstrual disorders & liver diseases. Bark is one of the constituents of a plaster used to promote maturation of boils and other malignant growths. The bark is recommended for the treatment of

menorrhagia and other uterine disorders. **As tooth brush** – The twigs are used as tooth brush. It is also sold in markets by the aboriginal people. They earn good amount from this. It serves as a additional source of income for them. **In venereal diseases** – Rhizomatous roots of *S. zeylanica* is used in venereal diseases such as syphilis & spermatorrhoea. Rhizome mixed with long pepper (*Piper longum*) crushed and extract is obtained. The extract is administered about 10 ml 3-4 times a day for 10-15 days. **For paralysis** – Tuberous root (rhizome) with long pepper (*Piper longum)* is crushed & extract is obtained. The extract is given about 5-10 ml 3-4 times a day for about a month for leprosy. **In throat ulcers** – Tuber paste is applied for throat ulcers. The leaves are eaten as vegetable by the rural indigenous people.

24. ***Solanum tuberosum*** L. 'Solanaceae"; POTATO;ALU. **Ethno-medicinal uses—In gastric ulcers, duodenal ulcers** — The tubers are washed nicely and the sprouts (the germinating young shoots) are removed. It is then rubbed to small pieces (grated) and squeezed in cheese-cloth to remove the juice. Half cup of the juice is administered orally before meals thrice daily for 10-15 days. It is very effective against duodenal and gastric ulcers. **In gastro-intestinal pains** —The potato tuber is also given in gastro-intestinal pains. It gives relief. Potato contains atropine- an analgesic substance that helps in easing gastro-intestinal pain. **In burns**- Crushed potato tuber juice is used for getting relief from burns. The juice combined with potato peels is considered a good remedy for topical burns.

25. Urginea indica (Roxb.)Kunth. "Liliaceae";JANGLI PIYAZ. **Ethno-medicinal uses— In abscess and boils**— The crushed bulb is tied on the abscess or the boils daily in the morning. The process is repeated till cured. The crushed bulb is tied on the abscess or the boils daily in the morning. The process is repeated till cured. **In leucorrhoea**—2-4 gm bulb powder is given once or twice for3-5 days to female in case of abnormal mucous discharge from the vagina (leucorrhoea). It gives relief to the patient. **In respiratory troubles**—two spoon bulb decoction is inhaled orally twice daily for 5-7 days for getting relief from respiratory troubles.

26. Zingiber officinale Rosc. " Zingiberaceae".ADRAK; ZINGER. **Ethno-medicinal uses–** The raw ginger is useful in condition of having lost the appetite for food (anorexia), indigestion (dyspepsia) & inflammations.The dry ginger is used in dropsy, otalgia, asthma, cough, diorrhoea, anorexia, cholera & inflammations. Eight to ten drops of the rhizome juice mixed with a teaspoon of honey, taken regularly for 3-4 days before bed time, cures throat infection. About 2 teaspoonful of the rhizome juice is taken with some sugar. It cures involuntary urination. The course is 3-times a day for 4-6 days. After each medication sufficient warm water is prescribed.Dry ginger is also used to prepare halwa for the mother just after delivery. It is called as Soth or Sonth. It serves as a very good agent for purification of new mothers' milk and gives strength to them. Dry ginger is also used in tea.**In abdominal pain**- One teaspoon rhizome powder is inhaled twice daily for 3-5 days orally with water. I t cures abdominal pain. **In vomiting** – One teaspoon rhizome powder with a bit salt is taken orally twice , it checks vomiting.

DISCUSSION

The dependency of human being on plants for their various needs such as food, fodder, shelter and even health care as medicines is known since prehistoric period.(Jain, S.K. & D.C. Pal1976[3]; Jain, S.K. 1995t ;Harshberger, J.W. 1896u ; Jain,S.K. & R.R. Rao1983v). The utility of plants as medicine by the ethnic tribes in different parts of India have been reported by many ethno-botanist(Jain, S.K.1963[1]; Jain, S.K. 1964[2]; Pal, D.C. 1973w ; Saxena et. al. 2001x ; Rai Choudhari et. al. 1980y ;Panigrahi, G. & S.K.Murthi 1989[1]R"; Maheshwari, J.K. 2003[11]). In spite of the fact that Jharkhand state is very rich in having ethno-botanically important plants in enormous numbers growing wildly in the Pat forests of the plateau,with 32 differennt ethnic tribe community residing in the state, no systematic or scientific study or documentation of traditional knowledge has yet been done. As regards to the ethno-botanical or ethno-medicinal documentation of the traditional folk lore system and horopathy (traditional ethnic tribal method of treatment with indigenous herbal medicinal plants) in Jharkhand state has not yet been done in a volumetric work. It is very scanty and fragmentary. Some work in this line have been done(Tarafdar, 1984a[12]; Tarafdar, 1984b[13]; Tarafdar 1983a[1]t ; Tarafdar 1983b[1]u ; Srivastava, D.K. & S.K. Verma 1981[1]v , Hembrom 1995[1]w ; Sahu, H.B. 2004[1]x ; Mahto R. 2010[1]y ; Gupta,

I.P.2011²R";Mahato, Subhash Chandra 2010[21]; Sahu, H.B.,S.L.Bondya, Jyoti Kumar & H.P.Sharma 2003[22];).

The present study has been chiefly confined with the documentation of traditional folk lore medicinal knowledge about the use of different tubers as medicine for the treatment of various ailments/diseases by the ethnic tribes of Khunti district in Jharkhand and collection of live forms. This type of work has not yet been done earlier either for any part of Jharkhand. Scientific identification and preservation of such tubers have been done following the standard methods (Jain, S.K. 1963[1]; Jain S.K. & Pal, D.C. 1976t ; Jain, S.K. 1995u). The medicinal properties of these tubers are still not known to the urban population and hence they do not utilize these tubers as medicine and go either to English medicines or modern system of treatments which are not only expensive but have the risk of having side effects. Thus these types of work needs urgent and wide publicity among the urban population so that they can also get acquainted about the folk lore medicines and start the same for their therapeutic needs. This practice will not only be helpful for the urban population but it will also be a mile stone for increasing the economy of the rural ethnic people by collecting and selling of such wild medicinally important plants.

Further it has also been noticed that due to different anthropogenic activities as indiscriminate exploitation of natural resources followed with cutting of forest wealth, erection of dams, establishment of industries and urbanization of the rural areas without providing any concrete substitute for the aboriginal population, this forest dependent population are being put to cross road and are being compel to live in a new environment which usually and in most cases is not hospitable for this mass. This mass feels uncomfortable and soon starts demolishing its own identity. Secondly due to these anthropogenic activities many ethno-medicinal plants are becoming either endangered or are being eradicated from the area or are at the verge of extinction. If this trend continues unabated there is every possibility for the extinction of the ethnic tribes communities from the area and at the same time the valuable treasure of ethno-medicinal plants will also be lost forever. As such there is an urgent need for the protection and preservation of the forest wealth not only in Khunti district but in whole of Jharkhand state to salvage the traditional folk lore medicines & horopathy, for the betterment of human society as a whole.

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